

✓ GB3: RISC-V Processor

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Introduction

Software applications reuse a small number of primitives (instructions) over and over again in time to implement a given computation. The domain of computer architecture deals with designing those primitives used by software and implementing them so that they are efficient according to some relevant measure such as time, energy, or implementation size. This IIA project will give you an opportunity to gain first-hand experience in modifying a popular computer architecture, the RV32I 32-bit integer variant of the RISC-V instruction set architecture (ISA).

The project will require you to first become familiar with the concepts of processor instructions, processor state, and processor microarchitecture. You will then explore how to implement basic hardware designs using the Verilog hardware description language and you will use this understanding to collaborate in a team to improve the performance of your microarchitectural implementation of the RV32I ISA.

RISC-V is pronounced “RISC five”.

Learning objectives

The project will introduce you to the concepts and tools for designing and modeling microprocessors, implementing hardware designs in FPGAs and evaluating their hardware resource usage, timing performance, and power dissipation.

The aim of the project is to understand how a processor is designed using Verilog, get first-hand experience implementing a baseline processor design on an FPGA, and modifying the processor to improve its hardware resource usage, timing performance, and/or power dissipation.

In this project, you will:

- learn the key concepts of computer architecture, including the concept of an instruction set architecture, processor state, and processor microarchitecture.
- learn how to use the Verilog hardware description language to design hardware.
- learn how to use open-source FPGA tools to synthesize and load Verilog hardware designs to the iCE40 FPGA.
- learn how to characterize the performance of a processor design.
- get experience optimising a processor design with respect to power consumption, timing performance, and resource usage.
- explore how to work effectively in a Team environment and identify appropriate roles for yourself

Intended Learning Outcomes

At the end of this project, you should be able to:

- create new hardware designs in the Verilog hardware description language.
- synthesise Verilog hardware designs for FPGAs using open-source FPGA tools.
- use your understanding of Verilog language and the related FPGA tools to modify a RISC-V processor design and evaluate it on an FPGA.
- propose and implement micro-architectural changes to a processor design to improve the performance and resource usage of the processor design.

Prerequisites (Assumptions)

This project assumes you to have some familiarity with the following concepts. If you are unfamiliar with any of the topics, please go through the resources recommended below.

- Basic concepts of computer architecture. If needed, you can go through the content for the Engineering Tripos Part IA (Digital Electronics and Information Processing), 3B2 (Integrated Digital Electronics) course before starting this project.
- Use of command line tools (e.g. basic shell scripts, creating Makefiles, and so on). If needed, you can watch [MIT's missing semester lecture on using the shell](#) for an introduction.

Session and Project Schedule

Compulsory sessions: Tuesdays 11-13, Fridays 9-11

Venue: EIETL (other than intro at LT6)

1. 15 May: Introduction, iCEBreaker 1 (basics)
2. 19 May: iCEBreaker 2 (Verilog testbenches)
3. 22 May: Picorv32 and PicoSoC
4. 26 May: Brainstorming
5. 29 May: Development 1
6. 2 June: Development 2
7. 5 June: Final presentations and demonstrations

Submissions

1. 21st May 4 pm (team): stopwatch.v, stopwatch_tb.v, an Excel file of data collected **[15 marks]**
2. 28th May 4 pm (team): plan, division of labour (roles and tasks) (2 pages A4 max.) **[15 marks]**
3. 11th June 4 pm
(individual): final report **[50 marks]**;
(group): all related design materials (diagrams, notes, code, lab books, etc.) **[no marks, supporting evidence for reports]**

Submit your work at [GB3 Moodle](#).

Guidelines for Plan

No more than 2 sides of A4. State your target(s) (performance, power and area) and the modification to be implemented. Propose a timeline and a balanced division of labour.

Tasks and roles: design, code, test, verify, measure, document

Rubric:

- Technical difficulty of the chosen modification
- Feasibility and time management
- Clarity of division labour to maximise output as a team

Guidelines for Final report

No more than 8 sides of A4, excluding appendices/references.

- Abstract (1 side)
- Summary of overall design decisions and outline of project management (1 side, possible team material)
- Description of individual design/code/test contribution (2-3 sides)
- Problems encountered by individual in development and their technical solutions (1 side)
- Conclusions, reflections and personal lessons learned (2 sides)
- Appendix: AI usage declaration (see note 2 below)
- List of references

Note: design materials are submitted as a team so no need to repeat them in the individual report

Note 2: Usage of AI is permitted in this project. If so, you will need to state clearly how AI was utilised during the planning, development and writing-up stages of the project. Also indicate if any materials are submitted verbatim (i.e. without any modifications) from AI, as they will be discredited in the grading process.

Session 1: iCEBreaker 1 (Basics)

Recommended tasks to complete.

- Complete the stopwatch workshop (<https://github.com/icebreaker-fpga/icebreaker-workshop/tree/master/stopwatch>)
 - You may skip chapter 2, as tools had been installed.
- Read the reports from the toolchain and note the FPGA resources
- Explore the available resources on the FPGA and test it on board

Session 2: iCEBreaker 2 (Testbench, PPA)

Please check in with Matthew when all members arrive at EIETL. And check out again when you finish.

Recommended tasks to complete. You will need to divide the tasks among team members.

- **Testing:** Create a testbench (`stopwatch_tb.v`) that will simulate the final (`stopwatch.v`) from session 1, as if you are doing manual checking on board.
- **Power:** Create a simple design with a single inverter (`inv_baseline.v`). Measure the power of this baseline against final stopwatch. Record results.
- **Performance:** Modify final stopwatch to output a 1 Hz square wave at any of the LEDs. Then use Picoscope to measure the output waveform. Take records and screenshots.
- **Area:** Study reports from FPGA tools and pick up resource usages.

Session 3: Picorv32 and PicoSoc

Recommended tasks to complete.

- Check out [Picorv32 & PicoSoC](#) source codes, then study the key source files (`picorv32.v`), (`picosoc/picosoc.v`), (`picosoc/icebreaker.v`).
 - Update as suggested [here](#).
 - **Draw a block diagram based on the Verilog codes.**
- Edit (`picosoc/Makefile`) as mentioned [here](#). Then program the PicoSoc onto the FPGA. Explore the demo functions (including UART) in this example.
- Simulate the design (Command: `make icebsim`)
- Download [blink.c](#) and try to program it (as the firmware) onto FPGA. Modify the C program to show your team number on the 7-segment display (`PMOD1A`) get a 1 Hz output on the LED using Picoscope.
- Measure the power of PicoSoc running (`blink.c`). How is it compared to the stopwatch and inverter baseline?

Session 4: Benchmarking & Brainstorming

Recommended tasks to complete.

- Continue with the block diagram if required. Make sure all members have a common understanding on the Picorv32 and PicoSoC.
- Using the while loop in main() of (`blink.c`) as an example, find the following
 - Number of instructions (N)
 - Cycles per instructions (CPI)
 - Clock period (T)

The product $N \times CPI \times T$ should match the execution time that you measure in Picoscope.

- Discuss as a team what target(s) (performance, area, power) and what modifications you are going to make.
- Meet Matthew and explain your plan by the end of the session.

✓ Session 5: Development 1

Restrictions on modifications

- Changes can only be made to hardware (*.v or *.sv) but not software (e.g. gcc compiler settings etc.)
- All firmwares (*.c) will be compiled using the standard `rv32im` architecture: (Makefile gcc options: `-march=rv32im`), without any other extensions.

Therefore, you may turn off compression support (`ENABLE_COMPRESSED = 0` in `picosoc.v`).

Each group will submit a **secret** benchmark (by 4th June 4 pm) for final evaluation, based on the following template.

```
unsigned char run_workload(void){
    // ...
    return x; // assume x is the final answer
}

void main(){
    // all required setup
    // ...
    unsigned char leds_value = 0x02;
    while (1) {
        // calculation that produces a unique answer
        reg_7seg = run_workload(); // display
        reg_leds = leds_value;
        leds_value = leds_value ^ 0x02; // toggle LED1
    }
}
```

This allows Picoscope to measure actual execution time through waveform at LED1.

An example of bubble sort as the workload is available [here \(bubble.c\)](#).

Session 6: Development 2

Team presentation at 12:30 p.m. to share main ideas and progress.

Session 7: Competition

You can download from [GB3 Moodle](#) the binary of RISC-V program (`secret*_fw.bin`).

Enter in this [spreadsheet](#):

1. Actual execution time on iCEBreaker against all 9 secret programs
2. The power of the board running your own secret program (please take away the baseline power)
3. The FPGA resource usages of your final design

✓ Hardware: iCEBreaker

This section covers the basic information of and set up for [iCEBreaker](#), the low cost, open-source educational field-programmable gate array (FPGA) development board.

The main motivating application of this board is for teaching the use of the open source FPGA design flow using Yosys, nextpnr, icestorm, iverilog, symbiflow and others.

✓ FPGA Development Toolchain

The OSS CAD suite toolchain is available on the Rocky Linux side of the EIETL PCs under path

```
/usr/local/apps/GB3-RISCV/oss-cad-suite
```

This command should start the OSS CAD environment

If you wish to install that onto your laptop, please follow this link <https://github.com/YosysHQ/oss-cad-suite-build>.

Software: OSS CAD Suite

Simulation (iVerilog): <https://steveicarus.github.io/iverilog/usage/simulation.html>

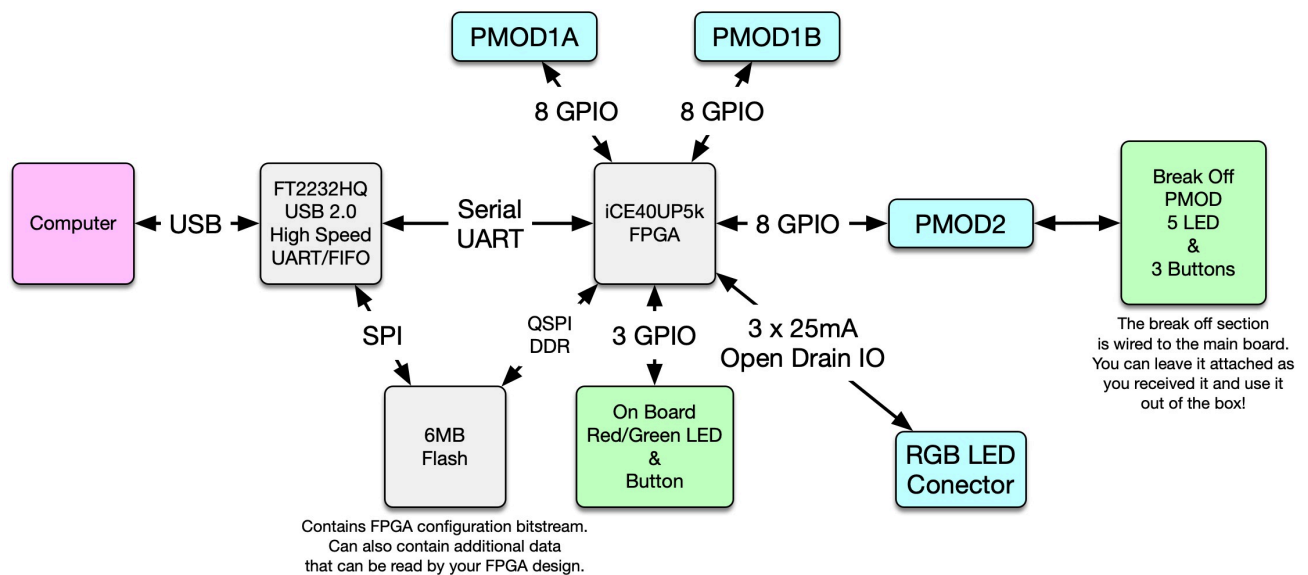
If you are new to Verilog testbench, this may be useful <https://fpgatutorial.com/how-to-write-a-basic-verilog-testbench/>.

iCEBreaker Hardware Specifications

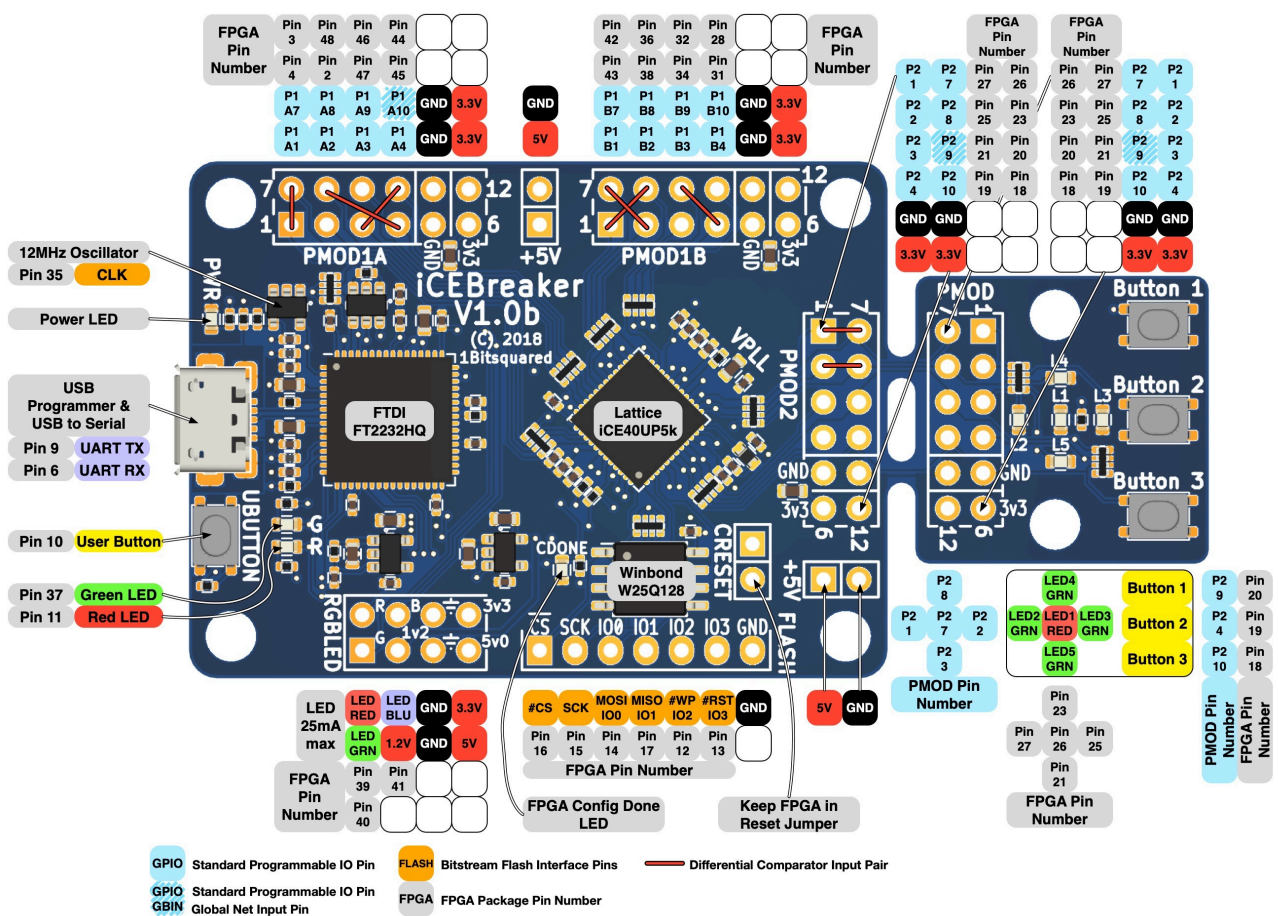
These are copied from iCEBreaker README, but only crucial resources for this project are included.

- iCE40UP5K in QFN48 (SG48) package
 - iCE40 UltraPlus 5K [Datasheet PDF Download](#)
 - **5280** Logic Cells (4-LUT + Carry + FF)
 - 128 KBit Dual-Port Block RAM
 - 1 MBit (128 KB) Single-Port RAM
 - PLL, Two SPI and two I2C hard IPs
 - Two internal oscillators (10 kHz and 48 MHz)
 - 8 DSPs (16x16 multiply + 32 bit accumulate)
 - 3x 24mA drive and 3x hard PWM IP
- QSPI-DDR-capable flash 128 MBit (16 MB)
 - The Winbond [W25Q128JVSIM](#) is used.
 - Projects may access the flash at the highest flash access speed possible.
- [FT232H](#) interface (microUSB plug)
 - programming using [iceprog](#) or [Dimond Programmer](#)
 - 12 MHz XTAL oscillator (shared with FPGA)
- 39 I/O capable pins:
 - 4 pins for config (SDI, SDO, SCK, CSB)
 - Either loading config from on board FLASH chip or provided through the FTDI chip with direct SRAM config.
 - 2 extra GPIO pins for QSPI
 - Together with the config pins, allows storage of additional data in the FLASH chip, and high speed QSPI DDR access. For example [picosoc](#) firmware.
 - 3 PINs for RGB LED (pin header)
 - 2 LEDs (one on output-only PLL pin)
 - 1 Clock pin (on PLL GBIN)
 - 1 UART Rx Pin via FTDI
 - 1 UART Tx Pin via FTDI
 - 1 Push Button
 - 16 PINs on dual PMOD
 - 8 PINs on single PMOD / snap-off section
- Break-off/Snap-off section (convertible to PMOD host / PMOD device)
 - 5 LEDs in similar arrangement to iCEstick
 - 3 Push Buttons
- Unpopulated parts shipped with the board
 - 3x Host PMOD (2x for dual PMOD port, 1x for snap-off section)
 - 1x Device PMOD (for other side of snap-off section)

iCEBreaker Diagrams



FPGA pins Layout



v1.1a schematic: <https://github.com/icebreaker-fpga/icebreaker/blob/master/hardware/v1.1a/icebreaker-sch.pdf>

Processor: RISC-V Picorv32 processor core / PicoSoC

In the [Picorv32 & PicoSoc Github](#), you can find the Verilog source codes, testbenches, firmwares (C programs) for PicoRV32 processor core that implements the RISC-V RV32IMC Instruction Set.

PicoSoC is a simple example SoC using PicoRV32, and it connects board I/O (i.e. UART, LEDs, PMODs) to the processor core.

Two important changes to the original source codes

1. Replace `icebreaker.pcf` with this [version](#). This will map the PMOD1A to the PicoSoC.
2. Replace `icebreaker.v` with this [version](#). This will add the 7-segment control and map it to the right GPIO address.

✓ Software: RISC-V Toolchain

We will consider the version installed on EIETL PC to be official RISC-V toolchain for this project.

Important: Change the variable `CROSS` in the Makefile of `picosoc` to use the install GNU toolchain:

```
CROSS=/usr/local/apps/GB3-RISCV/riscv32imc/bin/riscv32-unknown-elf-
```

Please make sure that your processor design will work with the software (firmware) built on the EIETL PC (if you choose to work on your laptops).

RISCV GNU toolchain (MacOS, Homebrew)

These are the instructions if you wish to install the toolchain on your own laptops.

Last tested: 16.11.2025 (Macbook Air M3, Tahoe 26.1) by Matthew Tang.

Add the homebrew tap from <https://github.com/riscv-software-src/homebrew-riscv> and install the tools.

```
brew tap riscv-software-src/riscv
brew install riscv-tools
brew edit riscv-gnu-toolchain
```

To install 32-bit cross-compiler, update the variables `args` to use `rv32i` architecture (32-bit integer) and `ilp32` (soft floating points)

```
args = [
  "--prefix=#{prefix}",
  "--with-cmodel=medany",
  "--with-arch=rv32i",
  "--with-abi=ilp32"
]
```

Then build the toolchain from source

```
brew reinstall --build-from-source riscv-gnu-toolchain
```

✓ Useful Utilities / Docking areas / Reminder for all

The following are useful commands or tools for the projects.

Linux Terminal Setup

Put the following lines into the `.bashrc_profile` file in your home directory. This will set an easy to use prompt on the terminal.

```
PS1='[\u@\h \w]$ '
alias ls='ls --color'
# settings for GB3 RISC-V
export SCREENDIR="$HOME/.screen/"
alias oss='source /usr/local/apps/GB3-RISCV/oss-cad-suite/environment'
alias iceuart='screen /dev/ttyUSB1 115200'
```

Serial port connection

Do this once: create a folder for temporary files for `screen`


```
mkdir -p $HOME/.screen/ && chmod 700 $HOME/.screen/
```

On a Rocky Linux terminal (EIETL/DPO), run the command

```
screen /dev/ttyUSB1 115200
```

if needed,

```
export SCREENDIR="$HOME/.screen/"
```

On a Mac terminal, run the command

```
screen /dev/tty.usbserial-ib9cp1CK1 115200
```

To terminate the connection, press Ctrl-A then k to kill.

Reference Books/Links

1. [Digital design and computer architecture \(RISC-V\)/ Sarah L. Harris, David Harris](#). Available across different colleges
Useful resources from this text: <https://pages.hmc.edu/harris/ddca/ddcarv.html>
2. [Computer organization and design RISC-V edition : the hardware software interface / David A. Patterson, John L. Hennessy](#). Online access via Cambridge library iDiscover